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RESEARCH ARTICLE



An analysis of the China–US trade war through the lens of the trade literature

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ABSTRACT

The current China–US trade war that started in early 2018 has been the largest of this kind in the global market in the past half century, if not longer. Many speculations about the reasons for and progress and potential implications of the trade war emerge. Countries must understand the reasons for the war to avoid future trade wars. Predicting what will happen in the near future and the related economic consequences are even more important for people (including businessmen and government policymakers) to prepare for them and make corresponding decisions. However, endeavouring to predict is a tough job. This paper tries to provide an unbiased analysis through the lens of the trade literature. That is, we want to ask how much we can understand the current trade war on the basis of the accumulated knowledge we can obtain from our profession. Related theories include imperfect competition, increasing returns, terms of trade argument, distributional effects and political economy argument.

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Introduction

China and the US are the two largest economies in the world. They are large not only in GDP but also among the largest in international trade and foreign direct investment (FDI). Undisputedly, the relationship between the two countries will have huge impacts on not only their economies but also the global economy. The effects, good or bad, are magnified by the global supply chains of production.

Unfortunately, a trade war is happening between these two countries, which started in early 2018. It started first by the US imposing an additional 25% of tariffs on many goods imported from China, and then China retaliated by imposing additional tariffs on some goods imported from the US. The trade war did not stop in the first round, and the two countries went to the second round of tariff escalations, and no sign of stopping is shown by the time of writing. This trade war has definitely been the largest in the global market in the past half century, if not longer. Many speculations about the reasons for and progress and potential implications of the trade war emerge.

Countries must understand the reasons for the war to avoid future trade wars. Predicting what will happen in the near future and the economic consequences are even more important for people (including businessmen and government policy-makers) to prepare for them and make corresponding decisions. However, endeavouring to predict is a tough job. This paper tries to provide an unbiased analysis through the lens of the trade literature. That is, we want to ask how much we can understand the current trade war on the basis of the accumulated knowledge we can obtain from our profession. To the best of our knowledge, this paper is the first one to present a systematic analysis of the current trade war in lieu of the existing trade literature.

For this purpose, we first review what has happened in reality. As early as August 2017, the US started to launch complaints about the unfair imports of steel and aluminium, but the accusation did not focus on China. Many countries are involved. However, since the mid of 2018, all the actions (raising tariffs) have been against China. China responded by raising tariffs on the imports from the US, which became a real trade war by any definition.

We then review the trade literature to search for theories and ideas that may help to explain trade wars or conflicts. Only a few studies in the literature are directly about trade wars, but familiar theories emerge, which can be adopted, with some reinterpretations, to explain trade wars. These theories include imperfect competition, increasing returns, terms of trade argument, distributional effects and political economy argument.

Finally, we apply those theories to the current situation of China and the US to determine which are practically relevant and which are not. The analysis corroborates that existing knowledge, based on the trade literature, does not offer a complete picture. However, suggesting complementary theories and arguments to fill the void is beyond the scope of this study.

US–China trade war

Outbreak of the trade war

As early as 18 August 2017, the Office of the United States Trade Representative (USTR) first obtained approval from the US government and initiated an investigation, based on Section 301 of the 1974 Trade Act, into China's laws, policies, practices and actions about technology transfer, intellectual property and innovation, as described in the USTR's notice on 24 August 2017.¹ The USTR issued a report on 27 March 2018, which focusses on unreasonable and discriminatory Chinese industrial practices and government interventions on technology transfer and licencing, intellectual property, investments and activities of the US companies.² In the report, the USTR proposed increasing tariff, WTO dispute settlement, and investment restrictions as appropriate ways to help address concerns on those issues and force China to change its unfair practices. On 20 June 2018, the USTR announced that it would adopt the means of increasing tariffs first.³

Specifically, on 15 June 2018, the USTR determined to impose an additional *ad valorem* duty of 25% on Chinese products with an approximate US\$34 billion annual

trade value, effective on 6 July 2018. Moreover, the USTR proposed another tariff increase with an approximate US\$16 billion annual trade value.

On 16 June 2018, as a response to the USTR's actions, China immediately retaliated with tit-for-tat tariffs by levying an additional *ad valorem* duty of 25% on approximately US\$50 billion annual trade value of the US goods exported to the Chinese market. On 17 September 2018, the USTR decided to impose an additional duty rate of 10% *ad valorem* with an approximate US\$200 billion annual trade value initially effective on 24 September 2018, and then the additional *ad valorem* duty will increase to 25% effective on 1 January 2019. To retaliate, China also imposed an additional *ad valorem* duty rate of 5% or 10% with a total approximate US\$60 billion annual trade value on the US goods. On 1 December 2018, China and the US came back to the conversation and reached a truce for 90 days. The US set the tariff level back to 10% on the US\$200 billion imports from China, and the latter agreed to increase imports from the former.

Official arguments for and against the trade war

According to the *Wall Street Journal* on 1 August 2018 (Davis and Wei 2018), the White House mainly has two purposes for the trade battle. The first purpose is to reduce the US\$376 billion annual trade deficit that the US had with China in 2017 by US\$200 billion, and the second is to force China to change its unequal industrial policies, which are clearly described in the USTR's report on 27 March 2018. The aim of the US actions is to stop unfair competition caused by Chinese industrial policies. In particular, as recorded in the White House remarks on 22 March and 29 May⁴, President Trump, on behalf of the US government, alleges that the Chinese government has unequal industrial policies toward enterprises with different ownerships for the goal of the *Made in China 2025* plan. Policies include subsidising favoured Chinese industries and requiring foreign firms to use technology transfer as an exchange for the Chinese market. Such policies of China threaten the global trading system to a great extent. The US government also criticises that China's WTO entry has brought negative influences on the US economy, including slower economy growth and lower employment rate because China exports much more competitive goods to the US market after 2001.

Furthermore, the USTR's 7 August report documents that President Trump directly suggested the USTR raise the tariff to 25% from the initial proposed rate of 10%. The same *Wall Street Journal* article (Davis and Wei 2018) infers that the US government might have two reasons for raising the tariff. One reason might be the dissatisfaction over the Chinese government's interventions in investment activities of US firms. For instance, a Chinese antitrust regulator denied US-based Qualcomm Inc.'s acquisition of Dutch chip maker NXP Semiconductors in July 2018. Another reason may be due to the consideration that the decline in the value of Chinese yuan over the past several months could actually help ease the tariff impacts given that a weaker Chinese yuan makes Chinese products more competitive in world markets. Thus, only a higher tariff could have more significant effects and therefore draw attention from the China side.

Regarding a series of tariff increase and false (according to China's point of view) accusations made by the US government, the information office of the State Council of the Chinese government issued an official report of *The Facts and China's Position on China-US Trade Friction*⁵ and asserted that the tariff increases abandon the basic standards for international communications, such as mutual respect and equal consultation. The accusations essentially reflect the unilateralism, protectionism and economic hegemonism of the US government. China claims that their markets and those of the US are highly complementary and that the divisions of the bilateral trade are principally based on the comparative advantages of the two countries. Specifically, products from the US are at the middle- and the high-end of the global value chain, and the US mainly exports capital and intermediate goods to China. Products from China are at the low- and the middle-end of the global value chain, and China mainly exports consumer goods and final products. Effectively, China-US trade has benefited both countries enormously for a long time. For instance, in China's export to and import from the US, the intra-industry trade of machines and computers ranks top in value amount, which reflects global production integration. On the one hand, the two countries' trade cooperation has promoted China's economic development and improved people's living quality. Under the economic globalisation framework, the cooperation has considerably contributed to integrating Chinese enterprises into the global production value chain and bringing a huge external benefit to China's economic growth. In terms of technological innovation and market management, the US firms that invest in China have also played a significant role in serving as good models for Chinese firms. On the other hand, entering the Chinese market and cross-border investments provide US firms remarkable business opportunities and hence make large contributions to economic growth, consumer welfare and economic structural upgrading of the US economy. For instance, China-US trade cooperation lowers the US inflation level because of the large amount of affordable and low-cost consumer goods imported from China. The trade cooperation also offers considerable employment opportunities in the US market.

As documented in the official report of *The Facts and China's Position on China-US Trade Friction*, in recent decades, but before the trade war, China and the US have shared the same perceptions on low tariff and strong intellectual property right (IPR) protection. After entering the WTO, China has fulfilled all tariff reduction commitments, and the total tariff level had dropped from 15.3% to 9.8% by the end of 2010. China has promoted trade and investment liberalisation by signing free trade agreements with other countries and given special preferential tariffs to the least developed countries. Furthermore, China's attitude toward the protection of IPRs has changed clearly. The 2000 amendment of the IPR law provides better IPR enforcement and a better environment for innovation by domestic and foreign firms in China. Meanwhile, *The Facts and China's Position on China-US Trade Friction* also claims that some official reports, for example, *Annual Business Environment Reports* issued by the China-US Chamber of Commerce from 2011 to 2018, positively affirmed China's intellectual property protection achievements in recent years. Moreover, as described in Guo et al. (2018), China has also given up 'secure and controllable' policy regarding the procurement process of information and communication technology to lower

technical barriers to trade in this area. However, the recent trade war has reflected the diverse views of the two countries on some issues.

First, the US government alleges that the large US trade deficit with China is caused by the unfair competition of Chinese firms supported by the Chinese government. China should remove import restrictions and buy more from the US. However, the Chinese government considers that the deficit is an inevitable result brought about by the US economic structure, for example, insufficient domestic saving. China also argues that the deficit is intrinsically determined by the comparative advantages and the international labour divisions of the two countries. Consequently, the slow growth and high unemployment of the US economy could not be primarily attributed to China's WTO entry and industrial policies.

Second, regarding technology transfer, the US government alleges that foreign firms are reluctantly forced to exchange technology for accessing the Chinese market under China's unfair industrial practice. On the contrary, the Chinese government asserts that the US firms take initiatives to make technology transfer decisions and that their actions are essentially in pursuit of their maximum benefits.

Third, as for intellectual property, the US government alleges that China not only imposes pressures on the US firms' technology transfers by means of foreign ownership restrictions but also helps conduct theft activities from the networks of the US firms to obtain intellectual property, trade secrets, or confidential business information. In connection with this affirmation, the Chinese government asserts that the slanderous accusation of the US government actually ignores the facts of dramatic progress and rapid development in the areas of innovation, science, and technology in China. In the past decades, China has established and gradually improved the intellectual property legal system. The US firms are provided with stronger protection and better legal environment than ever before.

Fourth, related to government interventions, the US government alleges that the Chinese government facilitates and subsidises Chinese firms in investments in and acquisitions of US companies to obtain cutting-edge technologies but imposes substantial restrictions on and intervenes in US firms' investments and activities in the Chinese market, which are unfair for the US firms. The Chinese government thinks that, on the basis of their needs to allocate resources and develop global markets efficiently, Chinese enterprises with high management abilities are encouraged to go abroad and carry out international economic cooperation, which is in line with the trends of economic globalisation and the WTO rules. Furthermore, as one of the means to deal with market failures and reduce imbalances in economic development, subsidy policies are actually widely used in many countries including the US, and the use of the subsidy policies by the Chinese government is consistent with the rules and obligations of WTO members. Furthermore, in the *Made in China 2025* plan, the Chinese government promises to provide equal opportunities to domestic and foreign firms.

Potential impacts on the markets

The outbreak of the trade war will definitely affect the two countries' economies and even the other countries' economies. Discussing the potential impacts on each country

and each market is not that difficult, but we must wait and see to verify any hypothetical analysis and prediction by data. We expect that many researchers would provide empirical analysis on this issue in the future.

However, the impacts of trade war on some markets are immediate. One such example is the financial market. Recently, Huang et al. (2018) have examined the market responses of firms to the trade war and have corroborated that US firms involved in trade with China tend to have poorer performance on stock and bond returns in the financial market around the outbreak of the trade war. Specifically, a 10% point increase in the ratio of a US firm's sales to China will lead to a 0.8% reduction of average cumulative returns in the stock market. Meanwhile, firms that are indirectly related to trade with China could also be influenced by the trade war through connections of global value chain engagement. Figure 1 shows public interests over trade war and S&P 500 index. Moreover, since the beginning of the trade war, the Shanghai Composite Stock Market Index has declined dramatically by approximately 20%, from average 3,300 in March 2018 to average 2,600 in January 2019. Distinct change could also be observed in the foreign exchange market and the exchange rate of Yuan per USD surged during the period. Figures 2 and 3 display the index changes in the Shanghai composite stock market and the exchange rate of Yuan per USD, respectively.

Furthermore, the trade war may exert persistent influences on the financial markets. Whenever the two governments make announcements on bilateral trade issue, investors in both countries respond promptly in the financial markets. For instance, when the US Commerce Department started to limit the US firms' business trade with China's ZTE Corporation on 16 April 2018, China stock market index declined immediately, especially for the stocks in the high-tech sector.

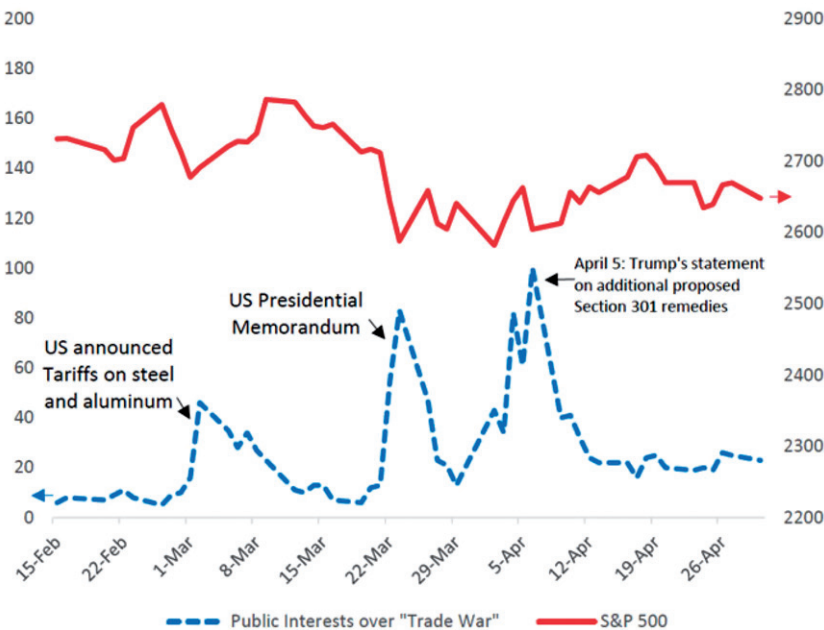


Figure 1. Public interests over trade war and stock returns. Data source: Huang et al. (2018).

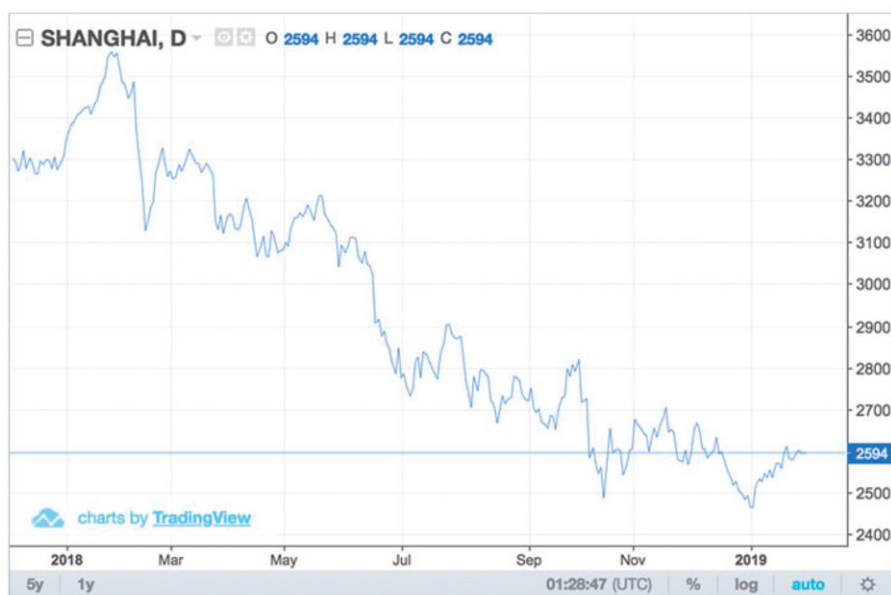


Figure 2. China Shanghai Composite Stock Market Index. Data source: TRADINGECONOMICS.com.

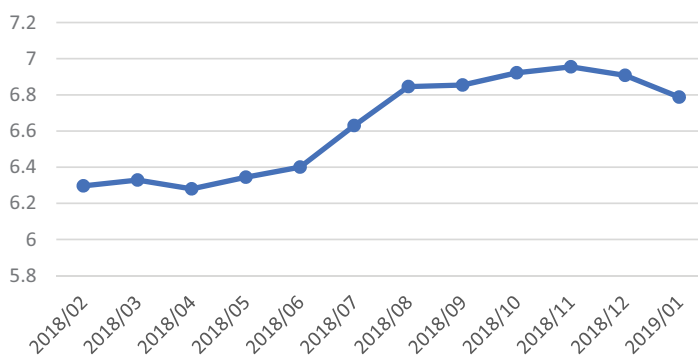


Figure 3. Exchange rate of Yuan per USD. Data source: US Bureau of Economic Analysis, <https://www.bea.gov>.

To sum up, the impacts of China–US trade war are potentially amplified in the financial markets in both countries by investors' anxious reactions towards the trade war.

Understanding from the lens of the trade literature

The belief that free international trade is beneficial for everyone is pervasively held by people all around the world. However, if that is the case, why is there a trade policy, or even trade war? The trade literature has provided several theoretical explanations for trade policy. The most popular explanations are imperfect competition, increasing returns, terms of trade argument, distributional effects, and other WTO and national security-related reasons.

Imperfect competition and increasing returns

In the world of David Ricardo, who assumes that market is efficient and that comparative advantages are the main drivers for international trade, free trade indeed benefits each country by expanding the production possibility frontier (PPF). Suppose that consumers consume wheat and rice in each country. If no trade emerges, then each country must produce wheat and rice to fulfil the needs of consumers. However, if trade emerges, then each country can concentrate on producing the product in which the country has a comparative advantage and then trade it for the other product that is produced by the other country with corresponding comparative advantage. Therefore, each country can produce more and consume more than the autarky case. The welfare is maximised when no trade barrier exists. Built on the Ricardian theory, Heckscher and Ohlin (HO) predicted that the patterns of production and trade are based on the factor endowment of a trading economy and that all countries benefit from trade. These are the two most fundamental arguments for free trade.

The Ricardian and HO models predict inter-industry trade that countries export products in industries with comparative advantages or abundant endowments. However, in recent decades, we have witnessed the change of trade patterns. One of the most important features is the emergence and growth of intra-industry trade. Instead of trading across industries, countries tend to trade similar products belonging to the same industry. For instance, the US imports cars such as Toyota from Japan, and the latter also imports cars such as General Motors from the former.

How to explain the intra-industry trade pattern? A theory including increasing returns and imperfect competition has been proposed, which is regarded as the New Trade Theory. Increasing returns to scale, or economies of scale, means that output increases by more than the proportional change in inputs. If one country specialises in certain varieties of a particular industry, then it may gain economies of scale from its specialisation that the marginal cost of production might reduce as the production increases. Firms from another country may specialise in other varieties of the industry. Consequently, intra-industry trade occurs. With increasing returns, new entrants cannot compete against the incumbent firms. Industries with economies of scale tend to have limited competition, leading to imperfect competition such as monopolistic competition.

These situations leave room for governments to help their firms increase their scale to compete with foreign firms and gain monopoly power to reap more benefits. As stated in the seminal work of Krugman (1987, 134), 'If increasing returns and imperfect competition are necessary parts of the explanation of international trade, however, we are living in a second-best world where government intervention can in principle improve on market outcomes'. Specifically, Krugman (1987) summarises two main arguments for trade policy: external economies and strategic trade policy.

External economies

The external economies' argument for trade policy is that the protection of industries with great externalities can be beneficial for the entire economy. With the existence of external economies, trade protection results in a larger domestic production of the industry and thus reduces the average cost of production. When the external

economies are sufficiently large, the benefit is more than compensating the cost of protection including the induced distortions.

One important form of external economies could arise from the accumulation of knowledge (Krugman, Obstfeld, and Melitz 2012) through learning-by-doing. When a firm refines its production techniques through experience, its production cost might reduce as the accumulation of production knowledge. This endeavour leads to the infant industry protection argument that suggests providing temporary protection for industries to help them gain experience. When external economies emerge, infant industries generally have a disadvantage in competing with their older competitors who have already attained a high level of accumulated output. Hence, the government should protect these industries until they can stand on their own. We now illustrate this argument with the learning curve. A learning curve describes the relationship that unit cost is decreasing with cumulative output.

Figure 4 illustrates the reason why temporary protection might help. Suppose that each country has one industry, and the learning curves are presented as L for one country and L^* for another. Suppose that the first country with learning curve L has somehow established the industry first and gained some experience that the unit cost has been reduced to C . The second country has a lower learning curve L^* , indicating that it could potentially be more productive in this industry. However, its initial unit cost is $C^* > C$, keeping it away from this industry. If the second country could provide some protection such that the cumulative output exceeds Q^* , its cost can be lower than that of the first country and then compete with the first country.

As for the China–US trade war, this argument is highly relevant. On the one hand, trade protection is higher in China. On the other hand, the Chinese government adopts subsidisation policy. Both are emphasised accusations by the US. We will analyse this point on the basis of the above theory in the below section.

Strategic trade policy

The main argument of strategic trade policy is that, with increasing returns and imperfect competition, protection may be able to make some industries earn returns higher than the opportunity costs of the resources they employ. In industries with oligopolistic

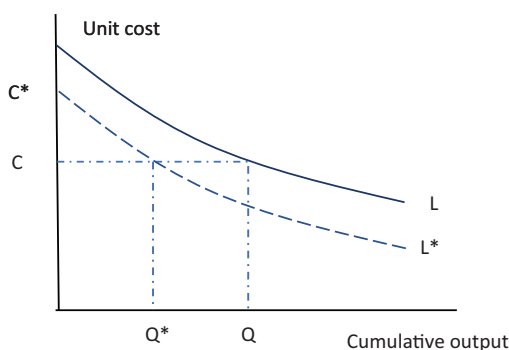


Figure 4. Learning curve and infant industry protection argument. Data source: Krugman, Obstfeld, and Melitz (2012).

competition, because firms earn positive rents, a government may adopt an appropriate trade policy to affect the strategic interaction between domestic and foreign rival firms such that the domestic firms’ profits can be increased at the expense of the foreign firms in quantity competition or consumers in price competition. Spencer and Brander (1983) and Brander and Spencer (1985) affirm that, under the right circumstances, government policies can deter foreign firms from competing for profitable markets. A simple numerical example by Krugman (1987) can help clarify the intuition.

Suppose that two firms produce aeroplanes for the global market, the Boeing in the US and the Airbus in Europe. Both firms choose to produce (P) or not to produce (N). Table 1 shows the hypothetical payoff matrix. Evidently, it does not have a unique outcome. Either Boeing produces and gains 100 payoffs, or Airbus produces and gains 100 payoffs. We suppose that Boeing has some advantages that it makes decision first; hence, the outcome would be Airbus exits from the market, and Boeing produces and earns high profits.

If the European government commits to provide a subsidy of 10 to Airbus, then no matter it produces or not, the payoff matrix is changed and presented in Table 2. The unique equilibrium is that Boeing stops producing, and Airbus produces with profit 110. Hence, a subsidy of 10 generates a profit of 110. This example clearly shows that strategic trade policy could potentially shift the profit from foreign firms (Boeing in this case) to domestic firms (Airbus in this case).

Table 1. Hypothetical payoff matrix.

		Airbus	
		P	N
Boeing	P	−5 −5	0 100
	N	100 0	0 0

Table 2. Hypothetical payoff matrix after European subsidy.

		Airbus	
		P	N
Boeing	P	5 −5	0 100
	N	110 0	0 0

If, theoretically, some room for trade policy to increase the overall welfare emerges, then why should we propose free trade across nations? Why should we bother to establish institutions such as WTO to promote the world to a free trade one? Krugman (1987) considers three main reasons.

First, a strategic trade policy could lead to retaliation and trade war. Suppose that two countries start at free trade. However, each country has the incentive to deviate to a high tariff. If its partner does not retaliate, then the country that unilaterally sets the optimal tariff could always benefit at the expense of its partner (see the example of Boeing and Airbus). In reality, countries indeed have incentives to deviate from free trade. According to Li and Qiu (2015), 1,162 trade disputes involving 110 countries ensued from 1995 to 2007, indicating that considerable deviations from free trade policy probably emerge.

However, the unilateral tariff will lead to the partner country to retaliate and result in a trade war. Ossa (2014) proves that, although each country can gain at the expense of other countries by unilaterally imposing optimal tariffs, when its partner retaliates, both will lose. This is a prisoners' dilemma. Free trade is a good equilibrium, but countries have incentives to deviate to the bad 'trade war' equilibrium. We can use a simple example presented in Table 3 to illustrate. Suppose that two countries, A and B, are in the world. Under free trade, both have payoffs 100. If A implements some protection policies while B stays free trade, then A can have a larger payoff 110, while B only gets 80. The total payoff shrinks because trade is less. Hence, the payoffs are (110, 80). The same case holds for B. However, if both protect their domestic market, then international trade shrinks, and producers and consumers suffer, leaving the payoffs to (50, 50). Evidently, both choosing free trade is the optimal results. However, each country has the incentive to deviate to protection, and the other country's best response is to retaliate, leading to bad equilibrium, which is a trade war.

To avoid that dilemma, Krugman (1987) suggests establishing rules of the game for policy that keep mutually harmful actions to a minimum and that these rules should be simple enough to be clearly defined. Free trade is such a simple rule. Consistent with Krugman's suggestions, in Ossa (2014)'s quantitative work, free trade is contended to be close to the efficiency frontier.

Second, the empirical difficulties in modelling imperfectly competitive market and its corresponding policies make free trade a simple enough policy to be enforceable.

Table 3. Free trade or trade war.

		Country B	
		Free Trade	Protection
Country A	Free Trade	100 100	110 80
	Protection	80 110	50 50

Modelling imperfect competition and proposing the right policies to enhance welfare are difficult. For instance, the effects of strategic trade policy in imperfectly competitive market depend crucially on how firms behave. Eaton and Grossman (1986) validate that, whether firms compete by setting prices or quantities in a duopoly model generates exactly the opposite optimal trade policy, export subsidisation, or taxation. The new entries after policy reforms and general equilibrium outcomes make the formulation and application of optimal trade policy more difficult.

Third, domestic politics matters. Trade policies might be influenced by domestic interest groups and used to raise the welfare of specific groups instead of the entire nation. This point is more related to political economy, which will be discussed later.

In sum, a strategic trade policy might be too difficult to formulate and apply. As an alternative, free trade policy might not be the optimal policy according to the theory but may be the best policy that the country is likely to obtain.

The strategic policy argument provides a controversial way of thinking the China–US war, i.e. whether the US initiated the trade war, or China did it. Two accusations from the US are particularly relevant, and they are the subsidies and entry regulations, which could function as *de facto* protective trade policies. The USTR investigation under Section 301 of China's laws, policies, practices or actions focusses much on technology transfer and intellectual property. It asserts that China's regulatory authorities force foreign firms that want to enter the Chinese market to transfer technology and intellectual property to Chinese firms for the purpose of attaining advanced technologies and commercial advantages. This type of non-tariff barrier, combined with the bad enforcement of Chinese IPR law, severely weakens the competitiveness of foreign firms, especially US firms with advanced technology. Distorted and unfavourable non-tariff practices could also move the profits of foreign firms to domestic firms. Moreover, the *Made in China 2025* plan, with the aim of dominating international markets in advanced manufacturing industries, might appear to be consistent with the theoretical finding of strategic trade policy's profit shifting incentives.

The US government also argues that the Chinese government has adopted distorted non-tariff trade measures, consisting of import quotas and licences, which lead to high trade deficits of the US with China. With these measures in place, together with Chinese subsidy policies, foreign firms are in disadvantage. The US asserts that this is unfair to other WTO members who firmly align with WTO trade obligations; therefore, the US must take actions to defend deserved rights and interests. Theoretically, distorted trade policies could result in retaliation and trade war, and both sides will finally lose in terms of social welfare because free trade is the most efficient and any deviation from free trade will ultimately be harmful. In fact, raising tariffs, along with WTO dispute settlement and investment restrictions, is one of the proposed means to address and fight against China's unfair trade practices by the US government as described in the USTR report of the Section 301 investigation of China's laws, policies, practices, or actions. Such type of retaliation could be likely to trigger more rounds of retaliations from each side and finally move toward a bad trade war equilibrium.

Strategic trade policy may be a pertinent argument for the US behaviour. According to the US accusations, the US appears to be retaliating against China's protective policies, instead of initiating the trade war. Thus, the key is to identify whether the

accusations are based on facts or not. We will discuss them in detail on the basis of existing research findings and results in the next section.

Terms of trade

In addition to the fundamental market structure argument, the traditional terms of trade argument also speak for trade policy. The terms of trade are the relative average price of a country's exports over that of imports. A rise of a country's terms of trade benefits that country in the sense that imports are relatively cheaper than exports, so that it can buy more imports given the level of exports. Therefore, this measure has been commonly used to capture gains from trade.

For a large country, which is able to affect the foreign exporters' prices, an import tariff has the potential to increase its terms of trade. Two opposite forces are induced by an import tariff. On the one hand, the tariff distorts the production and consumption of the country and hence reduces the aggregate welfare. On the other hand, when the country is large, the tariff lowers the price of imports, generating the terms of trade gain. When the terms of trade gain dominate the loss from distortion, the aggregate welfare of the country increases from the import tariff. The existence of this optimal tariff for large countries has been proved mathematically. Theoretically, large countries can always find an optimal tariff to increase their welfare.

However, this motive has not been actually used by governments as a justification for trade policy (Krugman, Obstfeld, and Melitz 2012). First of all, for small countries, only the negative distortion effect exists given that they are not likely to affect the world price, making them unable to have the terms of trade gain. Therefore, an import tariff always reduces a small country's welfare. The reason why big countries do not implement this policy in reality is similar to that of strategic trade policy. The implementation of optimal import tariff could probably bring retaliation from other large countries, ultimately leading to a bad trade-war equilibrium.

The above analysis also relates to the trade deficit argument that has been brought out from time to time by President Trump. As far as we know, the literature has not considered the trade deficit as a reason for trade war; instead, trade deficit is more like a result of an import tariff. Nevertheless, when a country (the US) improves its terms of trade by imposing an import tariff, the country will import less and may eventually reduce its trade deficit. However, because of retaliation, the strategy that uses import tariff to reduce trade deficit appears difficult to succeed. Stiglitz (2018) also asserts that the multilateral trade deficit matters, not the bilateral deficit, which is mainly determined by macroeconomics – the gap between domestic savings and investment, instead of US–China trade relations.

In addition, trade balance does not necessarily lead to higher social welfare. A recent paper by Guo et al. (2018) has examined the social welfare level of the US and China under different scenarios of unilateral US tariffs with balanced trade, US–China retaliatory tariff war with ongoing trade imbalance, and US–China retaliatory tariff war with balanced trade. Under the assumption that the US imposes 45% tariff, Guo et al. (2018) verify that, in all the scenarios, the US will undergo huge losses of social welfare,

while China might experience welfare gains or losses depending on the impact of the trade war.

In sum, we could not find convincing supports for the validation of President Trump's actions of raising tariffs to alleviate trade deficit and improve social welfare.

Distributional effects

When the market is imperfect, theoretically, trade policy could always improve welfare. However, even when the market is competitive, potential room for trade policy still emerges. For instance, the distributional effect has been treated as another 'reason' for trade policy. The distributional effect known as the famous Samuelson–Stolper theorem is one critical prediction in the HO model. It states that a rise in the relative price of a good will lead to a rise in the return to that factor that is used most intensively in the production of the good and, conversely, to a fall in the return to the other factor.

Developed countries are commonly more abundant in capital and hence tend to export capital-intensive products in the global market because its autarky relative price of capital-intensive to labour-intensive products is lower than the relative price in the global market. Therefore, trade will raise the relative price of capital-intensive products in developed countries and increase returns to capital while reducing returns to labour. These scenarios will enlarge the income gap between capital and labour in developed countries. The enlarged gap calls for trade policy to protect labour and reduce the income gap.

Theoretically, the distributional effect is not a legitimate argument for trade policy. When redistribution is efficient, international trade expands PPF. The winners' gains are large enough to compensate losses from losers, and everyone is better off. However, when redistribution, which is normally supposed to be the job of fiscal and monetary policies, does not work well, people who suffer from trade such as unskilled labour in advanced countries might wrongly blame international trade for its redistribution effect and vote for trade protection.

However, whether import competition from China generates benefits or losses to the US is still debated in academia. The seminal work by Autor, Dorn, and Hanson (2013) estimates that the rising imports from China explain 1/4 of the contemporaneous aggregate decline in the US manufacturing employment during 1990–2007. This estimated effect is non-trivial in reality and has been regarded as evidence of the need for trade protection. President Trump mentions that China's WTO entry has caused the US people suffering. However, in a recent study, Wang et al. (2018) confirm that imports from China will boost employment outside manufacturing through the downstream channel. Their results prove that the total impact of trading with China is positive to local employment and real wages. From the perspective of prices faced by US consumers, Amity et al. (2017) corroborate that China's entry into the WTO significantly reduced variety-adjusted US manufacturing price indices between 2000 and 2006, thus generated large welfare gains to the US. The academia has not formed a consensus on the aggregate effect of Chinese import competition on the US although we could witness some pieces of evidence on the distributional effect.

Political economy

In addition to the above-mentioned economic forces, political economic concerns are also important drivers for trade policies. A common but critical assumption of the political economy analysis is that the government is a benevolent servant of the national interest. However, the officials or leaders that run the government might not be a pure benevolent servant but might only care about their own benefits. Consequently, interest groups could lobby to affect the government's trade policies.

Grossman and Helpman (1994) affirm that, when politicians care about collecting political contributions and national welfare, interest groups will strategically choose the level of political contributions to influence the government's choice of trade policy. By applying this protection for sales model to a two-country framework, Grossman and Helpman (1995) demonstrate that interest groups could use campaign contributions to induce policy preferences in self-interested politicians, leading to trade wars in a non-cooperative international setting and trade talks in a cooperative international setting.

The interest groups lobbying explanation might not apply to the China–US trade war because Donald Trump self-raised his campaign funding; thus, theoretically, he is less likely to be influenced by interest groups.

Another influential strand of literature on political economy is that voters' preferences affect the trade policy formation as demonstrated by Mayer (1984) and subsequent researchers. Extensive literature on this topic, which focusses on politics, is available; thus, we are not going to discuss them in detail here.

Others

Other arguments for trade policy are based on the 'fairness' within the domain of WTO jurisdiction, such as the anti-dumping policy. On the basis of the WTO's rule, if a company exports a product at a price lower than the price it normally charges on its own home market, then it is regarded to be 'dumping' the product. This behaviour induces 'unfair' trade because firms are not competing with each other through market; instead, they are biased by some policies to have some extra advantages over other countries' exporters. The WTO agreement disciplines anti-dumping actions through regulations on how governments can or cannot react to dumping. Nowadays, the WTO dispute settlements of the US towards China's subsidy policies mainly focus on steel, aluminium and soda ash industries, and their low export prices have also raised many other countries' concerns of the dumping behaviour. However, anti-dumping is not the major focus of the US in this trade war, and we are not going to discuss it in detail.

Another important argument, with rather flexible interpretation, is the national security concerns. Some government intervention-related arguments in the USTR investigation under the Section 301 are actually based on the national security concerns.

Other arguments such as environmental protection or sanitary and phytosanitary concerns have not been used in this trade war; therefore, we skip them.

Further analysis

In this section, we provide a more in-depth analysis of two important arguments: subsidy and technology transfer.

Subsidy argument

Among all the arguments raised by the US, certain Chinese industrial policies, including the *Made in China 2025* plan, have been emphasised from time to time. Are industrial policies justifiable for the initiation of the trade war? From the theoretical analysis in the previous section, due to imperfect competition and increasing returns to scale, every country has incentives to deviate from free trade and introduce some protective policies. However, trading partners may retaliate. Therefore, industrial policies that build up *de facto* protection could potentially lead to a trade war.

Protective policies are pervasively used across countries, especially as a substitution for import tariffs after the GATT or WTO process of multilateral trade negotiations. As early as 1984, in the summary of trade policies of developed countries, Baldwin (1984) asserts that non-tariff barriers ‘have been used more extensively by governments to attain the protectionist goals formerly achieved with tariffs’. Since then, evidence on the substitution for import tariffs by other policies has been found in developing and developed countries including the US. These facts validate that many countries do have *de facto* ‘import tariffs’. However, for decades, we have not witnessed any country starting a large-scale retaliation and igniting a trade war. Not all protective policies lead to trade wars. This affirmation means that other factors behind the protective policies that are important for starting a trade war might emerge. Protective policies are not sufficient conditions for a trade war, or at least, policies that are not highly protective are not sufficient conditions. Thus, a simple accusation without quantifying the actual impacts of protective policies is not sufficient enough to start a trade war.

There could be two possible reasons for the US government to emphasise industrial policies in the accusation of China. First, it is merely an excuse, and second, the US truly believes that China’s industrial policies play an important role. If it is the latter case, the protection or discrimination induced by those policies must be large enough such that it makes the US worse off if it does not retaliate. In either case, we should learn more about China’s industrial policies.

What do industrial policies look like in China? Table 4 below, which is drawn from Aghion et al. (2015), shows the summary statistics of the different types of subsidies, including direct subsidies, tax holidays, interest rates and import tariffs, in China during 1998–2007. Several important features are presented in the table. First, subsidy policies are pervasive in China. Second, state-owned enterprises (SOEs) and foreign-invested enterprises (FIEs) are more preferred, while domestic private firms receive less protection. Between SOEs and FIEs, SOEs have higher coverage in subsidy but smaller coverage in tax holidays, and they have similar interest rates as FIEs. SOEs are substantially protected, and FIEs are protected by industrial policies in China.

If we look at subsidy distribution across industries (Table 5), then we could contend that significant variations in average subsidy during 1998–2007 across industries

Table 4. Industrial policy by ownership in China.

	1998	2000	2001	2002	2003	2004	2005	2007
All companies								
Percent of firms with subsidies	0.0937	0.11	0.115	0.129	0.138	0.151	0.137	0.124
Percent of firms with tax holidays	0.416	0.453	0.441	0.443	0.456	0.419	0.454	0.497
Ratio of interest payments to current liabilities	0.0557	0.0413	0.0366	0.034	0.0319	0.0268	0.0313	0.033
Average tariff on imports	19.48	18.68	13.84	13.58	12.23	10.91	10.17	10.12
SOEs only								
Percent of firms with subsidies	0.139	0.162	0.171	0.181	0.197	0.197	0.224	0.253
Percent of firms with tax holidays	0.306	0.355	0.334	0.343	0.365	0.337	0.367	0.455
Ratio of interest payments to current liabilities	0.0416	0.0288	0.0255	0.0238	0.0222	0.0184	0.0183	0.02
Average tariff on imports	19.81	19.11	13.76	13.48	12.05	11.01	10.24	10.24
Foreign firms only								
Percent of firms with subsidies	0.0678	0.0839	0.103	0.133	0.154	0.181	0.146	0.142
Percent of firms with tax holidays	0.54	0.591	0.572	0.585	0.593	0.577	0.598	0.608
Ratio of interest payments to current liabilities	0.0408	0.0282	0.0249	0.0219	0.0198	0.0164	0.0185	0.0198
Average tariff on imports	21.29	19.83	14.65	14.41	12.99	11.45	10.68	10.45
Domestic private firms only								
Percent of firms with subsidies	0.0835	0.105	0.107	0.119	0.126	0.138	0.131	0.116
Percent of firms with tax holidays	0.418	0.431	0.417	0.412	0.421	0.374	0.413	0.467
Ratio of interest payments to current liabilities	0.0668	0.0491	0.0424	0.0391	0.0365	0.0304	0.0356	0.0368
Average tariff on imports	18.65	18.14	13.58	13.33	12	10.74	10	10.02

Data source: Aghion et al. (2015).

Table 5. Industrial policy by sector in China.

Sector	Interest rate	Tariff	Subsidies	Tax holidays
Foodstuff	0.0424	21.67	0.109	0.476
Manufacture of beverages	0.0441	27.48	0.106	0.451
Manufacture of tobacco	0.0336	52.28	0.229	0.32
Manufacture of textiles	0.0357	14.39	0.12	0.444
Manufacture of textile wearing apparel, footwear	0.0256	20.32	0.101	0.492
Manufacture of leather, fur	0.0308	18.17	0.0959	0.486
Processing of timber, manufacture of wood, bamboo	0.0578	7.557	0.114	0.548
Manufacture of furniture	0.0397	8.776	0.0923	0.501
Manufacture of paper and paper products	0.0438	10.6	0.105	0.454
Manufacture of articles for culture, education and sport activity	0.023	11.99	0.126	0.474
Processing of petroleum, coking, processing of nuclear fuel	0.0391	6.046	0.106	0.388
Manufacture of raw chemical materials and chemical products	0.0391	9.513	0.145	0.452
Manufacture of medicines	0.0391	6.148	0.166	0.468
Manufacture of chemical fibres	0.0381	8.743	0.166	0.426
Manufacture of rubber	0.0376	15.66	0.116	0.455
Manufacture of plastics	0.0323	11.45	0.107	0.451
Manufacture of non-metallic mineral products	0.0462	12.38	0.139	0.445
Smelting and pressing of nonferrous metals	0.0367	6.193	0.109	0.413
Smelting and pressing of metals	0.0397	5.602	0.16	0.433
Manufacture of metal products	0.0293	12.15	0.107	0.432
Manufacture of special purpose machinery	0.0288	9.112	0.138	0.419
Manufacture of transport equipment	0.0289	17.57	0.15	0.413
Manufacture of electrical machinery and equipment	0.0266	11.67	0.144	0.423
Manufacture of communication equipment, computers and other electronic equipment	0.0182	7.081	0.155	0.538
Manufacture of measuring instruments and machinery for cultural activity and office work	0.0205	9.442	0.17	0.47
Manufacture of artwork and other manufacturing machinery for cultural activity and office work	0.0344	17.03	0.102	0.485
Manufacture of artwork and other manufacturing	0.0344	17.03	0.102	0.485

Data source: Aghion et al. (2015).

emerge. Given these basic facts, we now analyse China's subsidy policies under the theories we have summarised above.

First, Chinese government's subsidies on domestic firms is allegedly an unfair industrial policy. However, although the government appears to favour SOEs more, it also favours FIEs more than domestic private firms. That is, the discrimination is not totally against foreign firms in China.

Second, can we view the industry policies as consistent with infant industry argument given the fact that infant industry argument has been widely used for industrial and trade policies in developing countries? Table 4 shows that China's subsidy programme covers almost all industries. This practice does not appear to be supported by the need for infant industry protection. According to the USTR's 2017 Report to Congress on China's WTO Compliance, after China's 17-year WTO entry, China has not yet provided a complete list of subsidies that have been adopted by Chinese central and sub-central governments.⁶ The US once detected Chinese 'Famous Export Brand', 'World Top Brand', 'Export Bases' and 'Demonstration Bases' exports being related to subsidy programmes, covering textiles, agriculture, medical products, light industry, special chemical engineering, new materials, and hardware and building materials industries. Many of them do not belong to the infant industry category. Thus, one may conclude that infant industry theory cannot explain the government's subsidy policies.

Third, when externalities exist, private decisions in a free market are not able to reach the social welfare maximum point. Consequently, the government can intervene in the market to correct the market failure. Theoretically, when a government's mechanism is well designed, it is possible to achieve the outcome in which benefit covers the cost including distortions. Aghion et al. (2015) claim that well-designed industrial policies could promote competition and hence productivity and growth. Using Chinese data, they deduce that subsidy policies targeting at competitive industries, or policies that are diverse across firms within an industry, indeed foster competition. Thus, some subsidies are possibly designed for better efficiency.

Fourth, from the strategic trade policy point of view, if China actually builds up *de facto* trade barriers through domestic policies, the US will retaliate. In that case, the current trade war is supported by strategic trade policy. On a more specific view, Garred (2018) provides evidence that the distribution of China's export restrictions across industries increasingly resembles the inverse of its pre-WTO import tariff schedule. Garred (2018) also documents suggestive evidence, at 10% significance, on the *de facto* 'import tariff' in the sense that the export restrictions partly restored China's pre-WTO trade policy.

In sum, Chinese industrial policies are pervasive. While some policies are pro-competitive, certain policies that move in counter with tariff reductions also emerge, but their effects as *de facto* 'import tariff' on trade flows are not very strong. Therefore, the evidence is mixed with regard to whether Chinese industrial policies are critical in supporting the US initiation of the trade war. More direct evidence on the effects of Chinese industrial policies on China-US trade and on the US domestic economy is needed.

Technology transfer argument

A central accusation of China by the US is that US firms are forced to transfer their technologies to Chinese firms to gain access to the China market. In addition, many

Chinese firms ‘steal’ technology from US firms. Inward FDI has been regarded as an important source of economic growth through its positive spill-over effects. On the basis of this belief, international organisations and developed countries have put efforts to persuade developing countries to reduce restrictions on FDI for decades. However, the empirical evidence of spill-over effects is mixed due to the endogeneity concern that multinational firms endogenously make their investment decisions. They tend to choose to locate in areas with stronger IPR protection but will lower the benefit of spill-overs. Therefore, two questions must be addressed to examine the US accusation more carefully. First, we want to know whether technology transfer from US FDI in China can be carried out. Second, if the answer to the question is positive, then we aim to know whether the transfer is ‘forced’ or simply an optimal decision of US firms.

Recent research has inferred that, in general, positive spill-over effects from FDI on Chinese domestic firms are unlikely to emerge. Domestic firms are supposed to perform better in industries with more FDI presence if positive spill-overs exist. However, in recent empirical research, Lu, Tao, and Zhu (2017) identify a significantly negative spill-over effect of FDI on domestic Chinese firms. In industries with more FDI inflows induced by an arguably exogenous policy change, Chinese domestic firms tend to have lower productivity. Liu, Qiu, and Zhan (2018) also corroborate that domestic exporters in industries with higher FDI presence tend to export lower quality products. These negative effects do not support a positive technology transfer.

The above evidence speaks for the general FDI including greenfield and brownfield FDI and FDI from the US and from other countries. Will there be any different effects of FDI from different entry modes or from different countries? Jiang et al. (2018) find evidence of positive spill-over effects from international joint ventures (IJVs). When a foreign firm jointly sets up a JV with a Chinese partner firm, positive technology transfers to the JV and the Chinese partner firm emerge in terms of productivity and patents. Moreover, according to their estimates, JVs formed with US partners have higher productivity and generate higher positive spill-overs than those formed with non-US partners. One possible reason is that US firms are closer to the world technology frontier; thus, they transfer more or better technologies to Chinese firms. Although they find positive technology transfers from the US partners, whether the transfer is a ‘forced’ one or a natural consequence of cross-border investment remains to be unclear.

In general, FDI tends to have a net negative effect on local Chinese firms, i.e. the negative competition effect is larger than the positive technology spill-over effect, but IJVs formed with US partners are affirmed to have positive technology gains and positive spill-over to other domestic firms. Whether the technology transfer is forced or a natural result of economic cooperation or coordination is still an open question for research. In this sense, we could at least cautiously conclude that the US does not have sufficient evidence for the accusation of forced technology transfer or technology stealing.

Concluding remarks

The current China–US trade war is a surprise to most people. Hence, people become eager to learn why it happens, how it will evolve, and what the consequences we will face. Unlike most other articles that attempt to provide their own explanations, our

study has a less ambitious, but very important, task, that is, we try to see how much we can understand and explain the current trade war based on existing knowledge drawn from the trade literature.

We find that some theories can be adopted to explain trade war in general, directly or indirectly. These theories include imperfect competition and increasing returns, terms of trade argument, distributional effects, and political economy argument. However, when we apply them to the current situation of China and the US, we find that some are practically relevant and others are not. The analysis confirms that existing knowledge, based on the trade literature, does not offer a complete picture. Our analysis does provide a direction for future research to derive complementary theories and arguments to fill the void.

Notes

1. The Office of the USTR issued on 24 August 2017 'Notices: Initiation of Section 301 Investigation; Hearing; and Request for Public Comments: China's Acts, Policies, and Practices Related to Technology Transfer, Intellectual Property, and Innovation', which can be accessed in *Federal Register*, vol. 82, no. 163.
2. The US Executive Office of the President issued Presidential documents on 27 March 2018: 'Actions by the United States Related to the Section 301 Investigation Of China's Laws, Policies, Practices, or Actions Related to Technology Transfer, Intellectual Property, and Innovation', which can be accessed in *Federal Register*, vol. 83, no. 59.
3. The Office of the USTR issued on 20 June 2018 'Notice of Action And Request For Public Comment Concerning Proposed Determination Of Action Pursuant To Section 301: China's Acts, Policies, and Practices Related to Technology Transfer, Intellectual Property, and Innovation', which can be accessed in *Federal Register*, vol. 83, no. 119.
4. 'Remarks by President Trump at Signing of a Presidential Memorandum Targeting China's Economic Aggression', issued by the White House on 22 March 2018, retrieved from <https://www.whitehouse.gov/briefings-statements/remarks-president-trump-signing-presidential-memorandum-targeting-chinas-economic-aggression/>; and 'President Donald J. Trump Is Confronting China's Unfair Trade Policies', issued by the White House on 29 May 2018, retrieved from <https://www.whitehouse.gov/briefings-statements/president-donald-j-trump-confronting-chinas-unfair-trade-policies/>
5. *The Facts and China's Position on China-US Trade Friction* (which can be retrieved from <http://www.scio.gov.cn>) was released by the Information Office of the State Council, the People's Republic of China on 24 September 2018.
6. *2017 Report to Congress on China's WTO Compliance* (which can be retrieved from <https://ustr.gov>) was released by the USTR in January 2018.

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